A DIVISIBILITY SUBSEQUENCE WITH LARGE LOGARITHMIC COUNTING FUNCTION

Theorem 0.1. *Let*
$$A = \{a_1 < a_2 < \cdots\} \subset \{2, 3, \dots\}$$
be an infinite set of positive integers, listed in increasing order, and assume that A has positive lower logarithmic density, i.e.
$$\liminf_{x\to\infty} \frac{1}{\log x} \sum_{a_n < x} \frac{1}{a_n} > 0.$$
Then there exists an infinite subsequence
$$a_{n_1} < a_{n_2} < \cdots$$
such that
$$a_{n_i} \mid a_{n_{i+1}} \qquad (i \geq 1)$$
and
$$\limsup_{x\to\infty} \frac{1}{\log\log x} \#\{i : \ a_{n_i} < x\} \geq \limsup_{x\to\infty} \frac{1}{\log\log x} \sum_{a_n < x} \frac{1}{a_n \log a_n}.$$

Proof. Write
$$W_A(x) := \sum_{a_n < x} \frac{1}{a_n \log a_n}, \qquad \Lambda := \limsup_{x\to\infty} \frac{W_A(x)}{\log\log x}.$$
Note that
$$0 \leq W_A(x) \leq \sum_{2 \leq m < x} \frac{1}{m \log m} = \log\log x + O(1),$$
and therefore
$$0 \leq \Lambda \leq 1.$$
In particular, $\Lambda < \infty$.

We shall construct a random divisibility chain and show that, for at least one realization of that chain,
$$\limsup_{x\to\infty} \frac{1}{\log\log x} \#\big(\mathcal{C} \cap A \cap [2, x)\big) \geq \Lambda,$$
where $\mathcal{C}$ is the set of integers visited by the chain.

Step 1: a random divisibility chain. For each prime p, let $\xi_p \sim \mathrm{Exp}(1)$ be independent exponential random variables, and for $t > 1$ define
$$\alpha_p(t) := \left\lfloor \frac{\xi_p}{t \log p} \right\rfloor, \qquad N_t := \prod_p p^{\alpha_p(t)}.$$
For fixed $t > 1$,
$$\sum_p \mathbb{P}(\alpha_p(t) \geq 1) = \sum_p p^{-t} < \infty.$$
Hence, by the first Borel–Cantelli lemma, almost surely only finitely many $\alpha_p(t)$ are nonzero, and hence N_t is a well-defined positive integer.

To obtain a single probability-one event on which N_t is defined for every $t > 1$, let

$$\Omega_0 := \bigcap_{\substack{r\in\mathbb{Q}\\ r>1}} \{\#\{p:\ \xi_p \geq r\log p\} < \infty\}.$$

For each rational $r > 1$, the event inside the braces has probability 1 by the previous paragraph, so $\mathbb{P}(\Omega_0) = 1$. Now fix $\omega \in \Omega_0$ and $t > 1$. Choose a rational r with $1 < r < t$. Then

$$\{p:\ \xi_p(\omega) \geq t\log p\} \subseteq \{p:\ \xi_p(\omega) \geq r\log p\},$$

and the set on the right is finite by the definition of Ω_0. Hence only finitely many primes satisfy $\alpha_p(t,\omega) \geq 1$, so $N_t(\omega)$ is well-defined for every $t > 1$. Thus, after intersecting with the further probability-one events introduced below, we may and do work on a probability-one event on which N_t is defined for every $t > 1$.

If $1 < s < t$, then $\alpha_p(s) \geq \alpha_p(t)$ for every prime p, so

$$N_t \mid N_s.$$

Thus, as t decreases to 1, the distinct values taken by N_t form an increasing divisibility chain.

Let

$$T_{p,k} := \frac{\xi_p}{k\log p} \qquad (p \text{ prime},\ k \geq 1).$$

Among these, the values $T_{p,k} > 1$ are precisely the jump times of the process $t \mapsto N_t$ on $(1,\infty)$. For any two distinct pairs $(p,k) \neq (q,\ell)$, one has

$$\mathbb{P}(T_{p,k} = T_{q,\ell}) = 0.$$

Indeed, if $p \neq q$, then $T_{p,k}$ and $T_{q,\ell}$ are independent random variables with continuous distributions, so the probability that they are equal is 0. If $p = q$ and $k \neq \ell$, then

$$T_{p,k} = T_{p,\ell} \quad\Longrightarrow\quad \frac{\xi_p}{k\log p} = \frac{\xi_p}{\ell\log p} \quad\Longrightarrow\quad \xi_p = 0,$$

which again has probability 0. Since there are only countably many unordered pairs of distinct indices $((p,k),(q,\ell))$, a countable union bound shows that almost surely all jump times are distinct.

For later use, fix $u > 1$. Then

$$\sum_p \sum_{k\geq 1} \mathbb{P}(T_{p,k} > u) = \sum_p \sum_{k\geq 1} p^{-ku} = \sum_p \frac{p^{-u}}{1-p^{-u}} < \infty.$$

Hence, by the first Borel–Cantelli lemma, almost surely only finitely many jump times exceed u. Intersecting over rational $u > 1$, we may enlarge our probability-one event so that this holds simultaneously for every $u > 1$. On that event the process $t \mapsto N_t$ is piecewise constant on every interval $[u,\infty)$, with only finitely many jumps there. In particular, every visited state is attained on a nonempty interval of times, and for each $n \geq 2$ the event $\{n \in \mathcal{C}\}$ is measurable; indeed,

$$\{n \in \mathcal{C}\} = \bigcup_{\substack{r\in\mathbb{Q}\\ r>1}} \{N_r = n\}.$$

Moreover,

$$\sum_p \mathbb{P}(T_{p,1} > 1) = \sum_p \mathbb{P}(\xi_p > \log p) = \sum_p \frac{1}{p} = \infty.$$

By the second Borel–Cantelli lemma, almost surely infinitely many events $\{T_{p,1} > 1\}$ occur. Hence almost surely the process has infinitely many jumps on $(1,\infty)$, so the set

$$\mathcal{C} := \{N_t:\ t > 1\} \setminus \{1\}$$

is almost surely infinite and can be listed as

$$c_1 < c_2 < \cdots, \qquad c_i \mid c_{i+1} \quad (i \geq 1).$$

Step 2: the distribution of N_t.

For $n = \prod_p p^{v_p(n)}$ and $t > 1$, one has

$$\mathbb{P}\big(\alpha_p(t) = v\big) = p^{-vt}(1 - p^{-t}) \qquad (v \geq 0).$$

Let

$$E_y := \bigcap_{p \mid n} \{\alpha_p(t) = v_p(n)\} \cap \bigcap_{\substack{p \leq y \\ p \nmid n}} \{\alpha_p(t) = 0\},$$

where y runs over real numbers larger than every prime divisor of n. Then the events E_y decrease to $\{N_t = n\}$ as $y \to \infty$. Hence, by continuity from above and independence,

$$\mathbb{P}(N_t = n) = \lim_{y \to \infty} \mathbb{P}(E_y) = \prod_{p \mid n} p^{-v_p(n)t}(1 - p^{-t}) \prod_{p \nmid n} (1 - p^{-t}).$$

Since $t > 1$, the Euler product converges absolutely and

$$\prod_p (1 - p^{-t}) = \frac{1}{\zeta(t)}.$$

Therefore

$$\mathbb{P}(N_t = n) = \frac{1}{n^t \zeta(t)}. \tag{0.1}$$

Step 3: the probability that a given integer is visited. For $n \geq 2$, define

$$\rho(n) := \mathbb{P}(n \in \mathcal{C}).$$

Lemma 0.2. *For every $n \geq 2$,*

$$\rho(n) = \int_1^\infty \frac{1}{n^t \zeta(t)} \sum_{p \mid n} \frac{v_p(n) \log p}{1 - p^{-t}}\, dt.$$

Proof. Fix $n \geq 2$. By the observation added at the end of Step 1, on our probability-one event there are only finitely many jumps above each $u > 1$. Hence, if $n \in \mathcal{C}$, there is a well-defined first time $t_*(n) > 1$ at which the process enters the state n.

As t decreases, the process $t \mapsto N_t$ changes only at jump times, and at each jump exactly one prime exponent increases by 1 almost surely (because almost surely all jump times are distinct). Consequently every state can be entered at most once.

If $n \in \mathcal{C}$, let p be the prime whose coordinate jumps at time $t_*(n)$. Then necessarily $p \mid n$, and immediately before that jump the state is n/p. Thus the event $\{n \in \mathcal{C}\}$ is the disjoint union of the events $E_p(n)$ $(p \mid n)$, where $E_p(n)$ denotes the event that the process first enters the state n at a jump caused by the prime p. Therefore

$$\rho(n) = \sum_{p \mid n} \mathbb{P}(E_p(n)).$$

Fix $p \mid n$, and set

$$\tau_p := \frac{\xi_p}{v_p(n) \log p}.$$

Then τ_p has density

$$f_p(t) = v_p(n) \log p \cdot p^{-v_p(n)t} \mathbf{1}_{(0,\infty)}(t).$$

For almost every $t > 1$, conditional on $\tau_p = t$, the event $E_p(n)$ occurs if and only if

$$v_q(n)t\log q < \xi_q < (v_q(n)+1)t\log q \qquad (q \mid n,\ q \neq p),$$

and

$$\xi_q < t\log q \qquad (q \nmid n).$$

These conditions say precisely that just before time t, the exponent of each $q \neq p$ equals $v_q(n)$ if $q \mid n$, and equals 0 if $q \nmid n$. No extra condition is needed for p, because $\tau_p = t$ already forces the p-coordinate to jump from $v_p(n)-1$ to $v_p(n)$ at time t. Also, simultaneous jumps with some $q \neq p$ have conditional probability zero, since each ξ_q has a continuous distribution. Therefore, for almost every $t > 1$,

$$\mathbb{P}(E_p(n) \mid \tau_p = t) = \left(\prod_{\substack{q\mid n\\ q\neq p}} \left(q^{-v_q(n)t} - q^{-(v_q(n)+1)t}\right)\right)\left(\prod_{q\nmid n}(1-q^{-t})\right).$$

By the law of total probability with respect to the absolutely continuous random variable τ_p,

$$\mathbb{P}(E_p(n)) = \int_1^\infty \mathbb{P}(E_p(n) \mid \tau_p = t)\, f_p(t)\, dt,$$

and hence

$$\mathbb{P}(E_p(n)) = \int_1^\infty \left(\prod_{\substack{q\mid n\\ q\neq p}} \left(q^{-v_q(n)t} - q^{-(v_q(n)+1)t}\right)\right)\left(\prod_{q\nmid n}(1-q^{-t})\right) v_p(n)\log p\cdot p^{-v_p(n)t}\, dt.$$

For fixed $t > 1$, the infinite product over $q \nmid n$ is legitimate, since

$$\sum_{q\nmid n} q^{-t} < \infty,$$

and hence

$$\prod_{q\nmid n}(1-q^{-t})$$

is the probability of the independent event

$$\bigcap_{q\nmid n}\{\xi_q < t\log q\},$$

obtained as the limit over finite sets of primes.

Rewriting the finite product over $q \mid n,\ q \neq p$, we get

$$\mathbb{P}(E_p(n)) = \int_1^\infty v_p(n)\log p\cdot p^{-v_p(n)t}\prod_{\substack{q\mid n\\ q\neq p}} q^{-v_q(n)t}(1-q^{-t})\prod_{q\nmid n}(1-q^{-t})\, dt.$$

Summing over $p \mid n$, we obtain

$$\rho(n) = \int_1^\infty \left(\prod_{q\mid n} q^{-v_q(n)t}\prod_q (1-q^{-t})\right)\sum_{p\mid n}\frac{v_p(n)\log p}{1-p^{-t}}\, dt.$$

Since

$$\prod_{q\mid n} q^{-v_q(n)t} = n^{-t}, \qquad \prod_q (1-q^{-t}) = \frac{1}{\zeta(t)},$$

the stated formula follows. $\square$

Since

$$\sum_{p|n} v_p(n)\log p = \log n \qquad \text{and} \qquad \frac{1}{1-p^{-t}} \geq 1,$$

Lemma 0.2 immediately yields

$$\rho(n) \geq \log n \int_1^\infty \frac{dt}{n^t\zeta(t)}. \tag{0.2}$$

Lemma 0.3. *There exists an absolute constant $C > 0$ such that for every $n \geq 3$,*

$$\rho(n) \geq \frac{1}{n\log n} - \frac{C}{n(\log n)^2}.$$

Proof. Let $L := \log n$. From (0.2), after the change of variables $t = 1+u$,

$$\rho(n) \geq \frac{L}{n}\int_0^\infty \frac{e^{-Lu}}{\zeta(1+u)}\,du.$$

As $u \to 0^+$, one has

$$\frac{1}{\zeta(1+u)} = u + O(u^2).$$

Hence there exist absolute constants $B > 0$ and $u_0 \in (0,1]$ such that

$$\frac{1}{\zeta(1+u)} \geq u - Bu^2 \qquad (0 < u \leq u_0).$$

Now enlarge B if necessary so that the same inequality holds for all $u \in (0,1]$. Indeed, on the compact interval $[u_0, 1]$, the function $u \mapsto 1/\zeta(1+u)$ is continuous and strictly positive, whereas $u - Bu^2$ can be made uniformly smaller by increasing B. Therefore

$$\frac{1}{\zeta(1+u)} \geq u - Bu^2 \qquad (0 < u \leq 1).$$

It follows that

$$\rho(n) \geq \frac{L}{n}\int_0^1 e^{-Lu}(u - Bu^2)\,du.$$

Now

$$\int_0^1 ue^{-Lu}\,du = \frac{1-(L+1)e^{-L}}{L^2},$$

and

$$\int_0^1 u^2e^{-Lu}\,du \leq \int_0^\infty u^2e^{-Lu}\,du = \frac{2}{L^3}.$$

Thus

$$\rho(n) \geq \frac{1-(L+1)e^{-L}}{nL} - \frac{2B}{nL^2}.$$

Since $e^{-L} = 1/n$, this becomes

$$\rho(n) \geq \frac{1}{nL} - \frac{L+1}{n^2L} - \frac{2B}{nL^2}.$$

For $n \geq 3$, we have

$$\frac{L+1}{n^2L} \leq \frac{1}{nL^2}.$$

Therefore

$$\rho(n) \geq \frac{1}{nL} - \frac{2B+1}{nL^2}.$$

Thus the result holds with $C := 2B+1$. $\square$

Step 4: the expected number of visited elements of A. For a realization ω, let

$$C_A(x,\omega) := \#\big(\mathcal{C}(\omega) \cap A \cap [2,x)\big).$$

Then

$$\mathbb{E}\, C_A(x) = \sum_{a_n < x} \rho(a_n).$$

By Lemma 0.3,

$$\mathbb{E}\, C_A(x) \geq \sum_{\substack{a_n < x \\ a_n \geq 3}} \left(\frac{1}{a_n \log a_n} - \frac{C}{a_n (\log a_n)^2} \right).$$

Since the possible missing term $a_n = 2$ contributes at most the absolute constant

$$\frac{1}{2\log 2}$$

to $W_A(x)$, and since

$$\sum_{m \geq 3} \frac{1}{m(\log m)^2} < \infty,$$

there exists an absolute constant K such that

$$\mathbb{E}\, C_A(x) \geq W_A(x) - K \qquad (x \geq 3). \tag{0.3}$$

Step 5: a uniform second-moment bound. For each prime p, define

$$G_p := \left\lfloor \frac{\xi_p}{\log p} \right\rfloor.$$

Then G_p upper bounds the total number of jumps contributed by the prime p on the interval $t > 1$, and it is exactly that number except on the null event $\xi_p / \log p \in \mathbb{Z}$.

If $m \in \mathcal{C}$ and $m < x$, let $J(m)$ denote the unique jump at which the process first enters the state m. Distinct integers m give distinct jumps $J(m)$, since each jump produces exactly one new state. Moreover, the jump $J(m)$ multiplies the previous state by a prime divisor of m, hence by some prime $p < x$. Therefore the map

$$m \mapsto J(m)$$

embeds $\mathcal{C} \cap [2,x)$ into the collection of all jumps contributed by primes $p < x$. Since the total number of jumps contributed by a given prime p on $(1,\infty)$ is at most G_p, it follows that

$$\#\big(\mathcal{C} \cap [2,x)\big) \leq \sum_{p<x} G_p,$$

and in particular

$$C_A(x,\omega) \leq \sum_{p<x} G_p(\omega). \tag{0.4}$$

Next, for $k \geq 1$,

$$\mathbb{P}(G_p \geq k) = \mathbb{P}(\xi_p \geq k \log p) = p^{-k}.$$

Hence

$$\mathbb{E}\, G_p = \sum_{k \geq 1} \mathbb{P}(G_p \geq k) = \frac{1}{p-1},$$

and, using the standard tail-sum identity for nonnegative integer-valued random variables,

$$\mathbb{E}\, G_p^2 = \sum_{k\geq 1} (2k-1)\mathbb{P}(G_p \geq k) = \sum_{k \geq 1} (2k-1) p^{-k} \ll \frac{1}{p}.$$

Since the G_p are independent,

$$\mathbb{E}\Big(\sum_{p<x} G_p\Big)^2 = \sum_{p<x} \mathbb{E}\, G_p^2 + 2 \sum_{p<q<x} \mathbb{E}\, G_p \,\mathbb{E}\, G_q \ll \sum_{p<x} \frac{1}{p} + \left(\sum_{p<x} \frac{1}{p-1}\right)^2 .$$

Using the classical estimate

$$\sum_{p\le y} \frac{1}{p} = \log\log y + O(1),$$

together with

$$\frac{1}{p-1} = \frac{1}{p} + O\left(\frac{1}{p^2}\right),$$

we have

$$\sum_{p<x} \frac{1}{p-1} = \log\log x + O(1).$$

Therefore

$$\mathbb{E}\Big(\sum_{p<x} G_p\Big)^2 \ll (\log\log x)^2.$$

Together with (0.4), this gives

$$\sup_{x\ge e^e} \mathbb{E}\left(\frac{C_A(x)}{\log\log x}\right)^2 < \infty. \tag{0.5}$$

Set

$$Z_x := \frac{C_A(x)}{\log\log x} \qquad (x \ge e^e).$$

Then (0.5) says that $\sup_{x\ge e^e} \mathbb{E} Z_x^2 < \infty$. Hence for every $B > 0$,

$$\sup_{x\ge e^e} \mathbb{E}\big[Z_x \mathbf{1}_{\{Z_x>B\}}\big] \le \frac{1}{B} \sup_{x\ge e^e} \mathbb{E} Z_x^2,$$

because $Z_x \mathbf{1}_{\{Z_x>B\}} \le Z_x^2/B$. Therefore

$$\sup_{x\ge e^e} \mathbb{E}\big[Z_x \mathbf{1}_{\{Z_x>B\}}\big] \to 0 \qquad (B \to \infty),$$

so the family

$$\left\{ \frac{C_A(x)}{\log\log x} : x \ge e^e \right\}$$

is uniformly integrable.

We shall use the following elementary fact.

Claim 1. *Let $Y_1, Y_2, \ldots$ be nonnegative uniformly integrable random variables. Then*

$$\mathbb{E}\left[\limsup_{m\to\infty} Y_m\right] \ge \limsup_{m\to\infty} \mathbb{E}\, Y_m.$$

Proof. This is a special case of a reverse Fatou-type inequality under uniform integrability; see, for instance, [1, Corollary 4.2]. $\square$

Step 6: extracting one good realization. Choose a sequence $x_m \to \infty$ with $x_m \geq e^e$ for every m such that
$$\frac{W_A(x_m)}{\log\log x_m} \longrightarrow \Lambda.$$
This is possible by the definition of Λ. Set
$$Y_m := \frac{C_A(x_m)}{\log\log x_m}.$$
By (0.3),
$$\mathbb{E}\, Y_m \geq \frac{W_A(x_m)}{\log\log x_m} - \frac{K}{\log\log x_m},$$
so
$$\limsup_{m\to\infty} \mathbb{E}\, Y_m \geq \Lambda. \tag{0.6}$$

By (0.5), the family $\{Y_m\}$ is uniformly integrable. Therefore Claim 1, together with (0.6), yields
$$\mathbb{E}\left[\limsup_{m\to\infty} Y_m\right] \geq \limsup_{m\to\infty} \mathbb{E}\, Y_m \geq \Lambda.$$
Since $\Lambda < \infty$, the random variable $\limsup_{m\to\infty} Y_m$ cannot be strictly smaller than Λ almost surely; otherwise $\Lambda - \limsup_{m\to\infty} Y_m$ would be a positive integrable random variable with strictly positive expectation, contradicting the displayed inequality. Hence there exists a realization ω such that
$$\limsup_{m\to\infty} \frac{C_A(x_m,\omega)}{\log\log x_m} \geq \Lambda.$$
A fortiori,
$$\limsup_{x\to\infty} \frac{C_A(x,\omega)}{\log\log x} \geq \Lambda.$$

Since A has positive lower logarithmic density, we have $\Lambda > 0$. Indeed, let
$$\widetilde{S}_A(x) := \sum_{a_n \leq x} \frac{1}{a_n}.$$
Then
$$\liminf_{x\to\infty} \frac{\widetilde{S}_A(x)}{\log x} > 0,$$
because replacing $< x$ by $\leq x$ changes the sum by at most $1/x$. Hence there exist constants $\delta > 0$ and $X_0 \geq 2$ such that
$$\widetilde{S}_A(x) \geq \delta \log x \qquad (x \geq X_0).$$

Applying Stieltjes partial summation, and using $\widetilde{S}_A(2^-) = 0$, we obtain
$$\sum_{a_n \leq x} \frac{1}{a_n \log a_n} = \frac{\widetilde{S}_A(x)}{\log x} + \int_2^x \frac{\widetilde{S}_A(t)}{t(\log t)^2}\, dt.$$
Since replacing $\leq x$ by $< x$ changes the left-hand side by at most $1/(x\log x)$, it follows that
$$W_A(x) \geq \int_2^x \frac{\widetilde{S}_A(t)}{t(\log t)^2}\, dt - \frac{1}{x\log x}.$$
Hence, for $x \geq X_0$,
$$W_A(x) \geq \int_{X_0}^x \frac{\widetilde{S}_A(t)}{t(\log t)^2}\, dt - \frac{1}{2\log 2} \geq \delta \int_{X_0}^x \frac{dt}{t\log t} - \frac{1}{2\log 2} = \delta\bigl(\log\log x - \log\log X_0\bigr) - \frac{1}{2\log 2}.$$

Therefore
$$\Lambda = \limsup_{x\to\infty} \frac{W_A(x)}{\log\log x} \ge \delta > 0.$$

Consequently $C_A(x,\omega)$ is unbounded, so $\mathcal{C}(\omega)\cap A$ is infinite. List its elements increasingly as
$$a_{n_1} < a_{n_2} < \cdots .$$
Since $\mathcal{C}(\omega)$ itself is an increasing divisibility chain, we have
$$a_{n_i} \mid a_{n_{i+1}} \qquad (i \ge 1).$$
Finally,
$$C_A(x,\omega) = \#\{i:\ a_{n_i} < x\},$$
and hence
$$\limsup_{x\to\infty} \frac{1}{\log\log x}\#\{i:\ a_{n_i} < x\} = \limsup_{x\to\infty} \frac{C_A(x,\omega)}{\log\log x} \ge \Lambda = \limsup_{x\to\infty} \frac{1}{\log\log x} \sum_{a_n<x} \frac{1}{a_n \log a_n}.$$
This completes the proof. □

Remark 0.4. The argument above actually uses only the positivity of
$$\Lambda = \limsup_{x\to\infty} \frac{1}{\log\log x} \sum_{a_n<x} \frac{1}{a_n \log a_n}.$$
Thus the same proof gives the conclusion under the weaker hypothesis $\Lambda > 0$; the assumption of positive lower logarithmic density is used only to guarantee that $\Lambda > 0$.

References

[1] Takashi Kamihigashi, A generalization of Fatou's lemma for extended real-valued functions on σ-finite measure spaces: with an application to infinite-horizon optimization in discrete time, *Journal of Inequalities and Applications* **2017** (2017), Article 24. DOI: 10.1186/s13660-016-1288-5.